

Di-Ao Liu, Ph.D.

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Treat an illness before it occurs

Education

University of Pennsylvania

Philadelphia, PA

Ph.D., Biological Science (age 27)

July 2018 – May 2024

- Thesis title: “Molecular mechanism of exosome secretion”
- Advisor: Wei Guo, Hirsch Family President's Distinguished Professor, Department of Biology
- Committee: Dr. Michael Lampson, Dr. Michael Marks, Dr. Serge Fuchs, and Dr. Tatyana Svitkina
- Focused on understanding the mechanism of exosome secretion and cancer development
- Defined the exosome secretion fate determination by PI(4)P and the exocyst complex.

Johns Hopkins University

Baltimore, MD

Visiting Student, Biochemistry

July 2017 – Aug 2017

Sun Yat-Sen University

Guangzhou, China

B.S., Life Science (age 21)

Sep 2014 – Jun 2018

- Thesis: SHISA family members are new autophagy receptors for immune regulation. (This work later leads to published work in Nature Microbiology on SHISA9)
- Yat-Sen Honor School Program (Top 0.5%)
- Honored undergraduate student of Biology and Yat-sen School
- GPA: 3.97/4 Rank in major: 2/88

Research Experience

Meyer Cancer Center, Weill Cornell Medicine

New York, NY

Postdoctoral Fellow, David Lyden's Lab

July 2024 – Present

- Co-advised by Dr. Haiying Zhang and Dr. David Lyden (Stavros S. Niarchos Professor)
- Exploring extracellular vesicle DNA biology and exosome biogenesis
- Focused research on T cell EV-DNA biology and tumor cell EV-ecDNA biology
- Establishing CRISPRi-Screening based extracellular vesicle and particle biogenesis studies

Department of Biology, University of Pennsylvania

Boston, MA

Graduate Researcher, Wei Guo's Lab, Extracellular vesicle biology

Mar 2019 – May 2024

Penn Medical School

Boston, MA

Rotation Graduate Researcher, Hongjun Song's Lab, scRNAseq of GBM organoid

Jan 2019 – Mar 2019

Department of Biology, University of Pennsylvania

Boston, MA

Rotation Graduate Researcher, Michael Lampson's Lab, Optogenetics

Oct 2018 – Dec 2018

Sun Yat-Sen University

Guangzhou, China

Undergraduate Researcher, Jun Cui's Lab

2015 – 2018

Publications

† Corresponding author; * Co-first author

Selected Publications

1. Hu, M. *, **Liu, D. A.***, Wortzel, I., Collier, P., Nelson, T. M., Foon, J., . . . Lyden, D. Activated T cell extracellular vesicle DNA transfer enhances antigen presentation and anti-tumor immunity. **Cancer Cell**. 44, 965-982 e912 (2026). <https://doi.org/10.1016/j.ccell.2026.03.023>
Featured in Nature Review Immunology, Cancer Cell, WCM News and other media
2. **Liu, D. A.**, Tao, K., Wu, B., Yu, Z., Szczepaniak, M., Rames, M., . . . Guo, W. A phosphoinositide switch mediates exocyst recruitment to multivesicular endosomes for exosome secretion. **Nature Communications** 14, 6883 (2023). <https://doi.org/10.1038/s41467-023-42661-0>
3. Wu, B. *, **Liu, D. A.***, Guan, L. *, Myint, P. K., Chin, L., Dang, H., . . . Guo, W. Stiff matrix induces exosome secretion and promotes tumor growth. **Nature Cell Biol** 25, 415-424 (2023). <https://doi.org/10.1038/s41556-023-01092-1>
Featured in NCB News & Views, Penn Today and other media
4. Di-Ao Liu† and Wei Guo†. The membrane origin of extracellular vesicle (In preparation).

Other Publications

5. Parihar, K., **Liu, D. A.**, Hassan, G., Issadore, D. A., Janmey, P. A., Weaver, V. M., Guo, W. & Radhakrishnan, R. Mechanical regulation of extracellular vesicle activity during tumour progression. **Nat Biomed Eng** 9, 1202-1221 (2025). <https://doi.org/10.1038/s41551-025-01446-0>
6. Jacob F, Salinas RD, Zhang DY, Nguyen PTT, Schnoll JG, Wong SZH, Thokala R, Sheikh S, Saxena D, Prokop S, **Liu D.A.**, Qian X, Petrov D, Lucas T, Chen HI, Dorsey JF, Christian KM, Binder ZA, Nasrallah M, Brem S, O'Rourke DM, Ming GL, Song H. A Patient-Derived Glioblastoma Organoid Model and Biobank Recapitulates Inter- and Intra-tumoral Heterogeneity. **Cell** 188-204 e122, (2020). <https://doi.org/10.1016/j.cell.2019.11.036>
7. Hou, J. Mei, K. Wang, D. Ke, S. Chen, X. Shang, J. Li, G. Gao, Y. Xiong, H. Zhang, H. Chen, L. Zhang, W. Deng, Y. Hong, X. **Liu, D. A.** Hu, T. Guo, W. Zhan, Y. Y. TGM1/3-mediated transamidation of Exo70 promotes tumor metastasis upon LKB1 inactivation. **Cell Rep** 43, 114604 (2024). <https://doi.org/10.1016/j.celrep.2024.114604>
8. Shen, H. Atiyas, Y. Yang, Z. Lin, A. A. Yang, J. **Liu, D.A.** Park, J. Guo, W. Issadore, D. A. Ultrasensitive quantification of PD-L1+ extracellular vesicles in melanoma patient plasma using a parallelized high throughput droplet digital assay. **Lab Chip** 24, 3403-3411 (2024). <https://doi.org/10.1039/d4lc00331d>
9. K Mei, **Liu, D.A.** W Guo. Mei, K., Liu, D. A. & Guo, W. Determine the Function of the Exocyst in Vesicle Tethering by Ectopic Targeting. **Methods Mol Biol** 2473, 65-77 (2022). https://doi.org/10.1007/978-1-0716-2209-4_6
10. Xie, W., Jin, S., Wu, Y., Xian, H., Tian, S., **Liu, D. A.**, Guo, Z. & Cui, J. Auto-ubiquitination of NEDD4-1 Recruits USP13 to Facilitate Autophagy through Deubiquitinating VPS34. **Cell Rep** 30, 2807-2819

e2804 (2020). <https://doi.org/10.1016/j.celrep.2020.01.088>

11. Jin, S., Zhang, X., Miao, Y., Liang, P., Zhu, K., She, Y., Wu, Y., **Liu, D. A.**, Huang, J., Ren, J. & Cui, J. m(6)A RNA modification controls autophagy through upregulating ULK1 protein abundance. **Cell Res** 28, 955-957 (2018). <https://doi.org/10.1038/s41422-018-0069-8>

Skills

Biological Experiments: Exosome preparations, LNP, Genetic vector cloning, Whole Genome Sequencing, Flow Cytometry, Confocal Imaging, Electronic microscopy, Exosome DNA biology, Particle analysis technology, Mouse models, CRISPRi genetic screening, ELISA, scRNAseq.

Adaptability: Fast understanding of designs and protocols, optimization of protocols.

Data Analysis: R Code, Python Code, Basic Ai Agent Usage.

Office Presentation: Adobe Illustrator (Nature Review Level Figures), Microsoft Word/EXCEL/POWERPOINT.

Presentations

WCM Pediatric Postdoc Research Day 2026	New York, NY
Multicellular Oncogene Persistence Driven by Escaped Extracellular ecDNA	2026
WCM Pediatric Postdoc Research Day 2025	New York, NY
The biogenesis of exomere	2025
ASCB Cell Bio Consortium 2022	Washington, DC
Exocyst controls exosome secretion by phosphoinositides switch	2022

Awards

National Outstanding Abroad Student Fellowship	2024
Penn Bio PhD Fellowship	2018
iGEM 2016 Gold Prize, iGEM	2016
National Scholarship of China , the Ministry of Education	2014 – 2017
Yat-Sen Scholarship , Sun Yat-Sen University	2015 – 2018
Yat-Sen Honor School Program , Sun Yat-Sen University	2015 – 2018
National Olympiad of Biology in Provinces, First Prize	2012

Professional Experience

Service

Associated Faculty Member, Faculty Opinions 2020 – Present

Organizing Committee Member, CCiC (China College iGEM Consortium) 2016 2016

Mentoring

BIOL course instructor, Department of Biology, University of Pennsylvania 2019

Students Supervised

Predoctoral Students: Yuhao Wan, Yao Fu, Leon Li.

MD-PhD Students: Andrew Chang

References

Dr. David C. Lyden, Postdoctoral Co-Advisor dcl2001@med.cornell.edu

Stavros S. Niarchos Professor, Pediatrics, Weill Cornell Medicine

Dr. Haiying Zhang, Postdoctoral Co-Advisor haz2005@med.cornell.edu

Assistant Professor of Pediatrics, Weill Cornell Medicine

Dr. Wei Guo, Thesis Advisor guowei@sas.upenn.edu

Pew Scholar, Hirsch Family President's Distinguished Professor, University of Pennsylvania